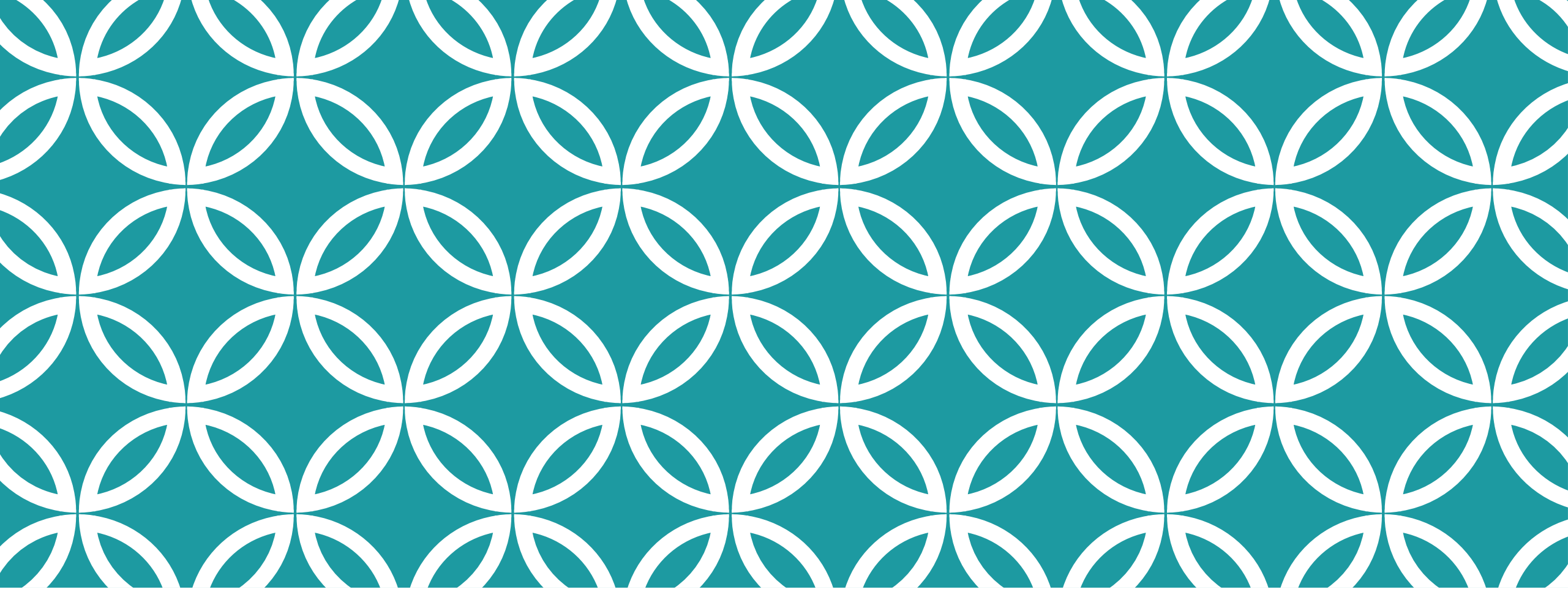


STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

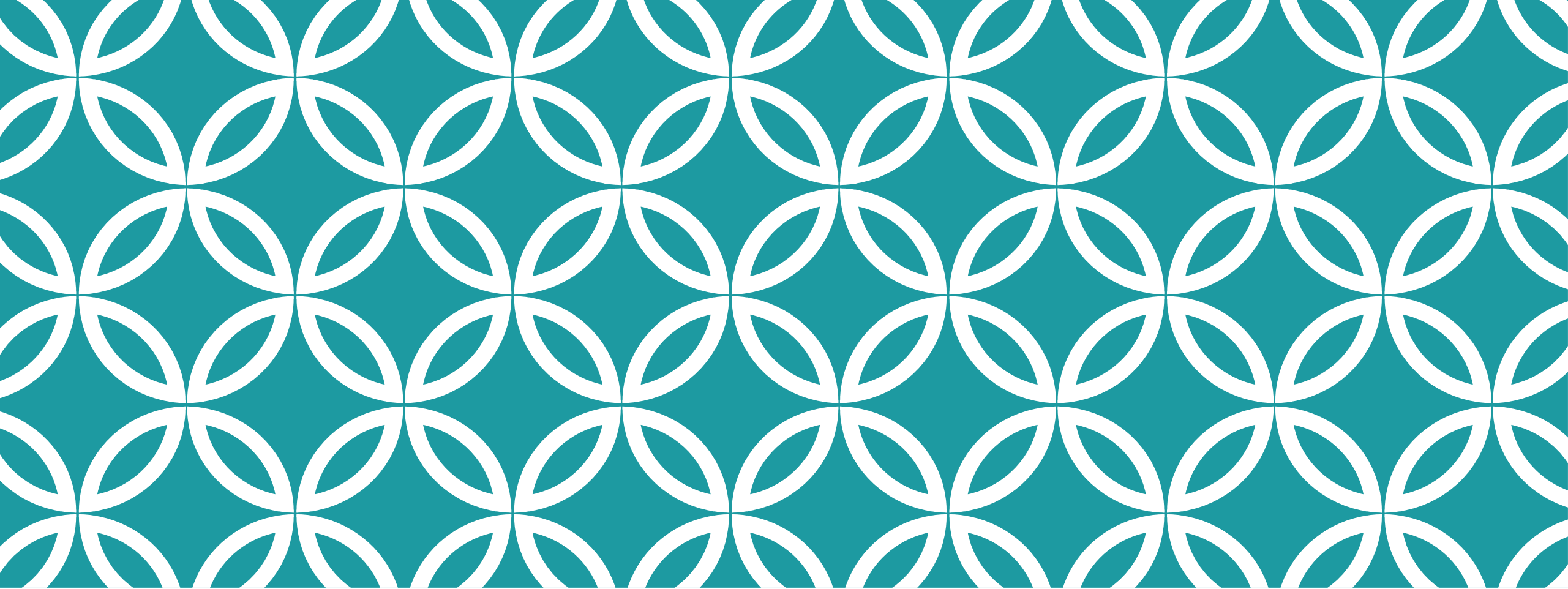
Week-3: Exercises-Data Collection and analysis and Concepts Variables and Measurement

Shivakumar Jolad
Associate
Professor, FLAME
University, Pune



DISCUSSION FORUM- STUDENT RESPONSES

IN CLASS DISCUSSION



CENSUS DATA |

Census of India 2011 Houselisting and Housing Census Schedule

Confidential
when filled

Page No.

SIDE-A

Start Here	Location Particulars	State/UT	District	Tahsil/Taluk/ P.S./Dev. Block/ Circle/Mandal	Town/ Village	Ward Code No. (only for Town)	Houselisting Block No.																											
Line number	Building number (Municipal or local authority or census number)	Census house number	Predominant material of floor, wall and roof of the census house (Give code number from the respective lists below)	Ascertain use of Census house (Write the actual use and then choose the appropriate code from the list below and record the same in the box at the left hand side of the column)	Household number (Give separate serial number to each household and write '999' for every institutional household)	Total number of persons normally residing in this household	Name of the head of the household	Sex: Male-1/ Female-2	If SC* or ST* or Other: SC-1/ ST-2/ Other-3	Ownership status of this house: Owned-1/ Rented-2/ Any other-3	Number of dwelling rooms* exclusively in possession of this household (Record 0,1,2,3...)	Number of married couple(s) living in this household (Record 0,1,2,3...)	Main source of drinking water: (Give Code number from the list below)	Availability of drinking water source: Within the premises-1/ Near the premises-2/ Away-3	Main source of lighting: Electricity-1, Kerosene-2, Solar-3, Other oil-4, Any other-5, No lighting-6	Latrine within the premises: Yes-1/ No-2	If '1' in col. 22, give Code from 1 to 8; if '2' in col. 22, give Code 9 or 0 from the list below	Waste water outlet connected to: Closed drainage-1/Open drainage-2/ No drainage-3	Bathing facility available within the premises: Yes: Bathroom-1, Enclosure without roof-2/No-3	Availability of kitchen: (Give Code number from the list below)	Fuel used for cooking: (Give Code number from the list below)	Radio/ Transistor: Yes-1/ No-2	Television: Yes-1/ No-2	Computer/Laptop: Yes: With Internet-1, Without Internet-2/ No-3	Telephone/Mobile phone: Yes: Landline only-1, Mobile only-2, Both-3/No-4	Bicycle: Yes-1/ No-2	Scooter/ Motor Cycle/ Moped: Yes-1/ No-2	Car/ Jeep/ Van: Yes-1/ No-2	Availing Banking Services: Yes-1/ No-2					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35

Note: * for col. 15 Scheduled Caste can be only from Hindus, Sikhs and Buddhists and not from any other religion. Scheduled Tribe can be from any religion. # for col. 17 Do not include kitchen, bathroom, latrine, store room, passageway and verandah. Include living room, bedroom, dining room, drawing room, study room, servant's room and other similar rooms.	Count the number of entries and give total	Total					
4 Floor Mud 1 Wood/ bamboo 2 Burnt brick 3 Stone 4 Cement 5 Mosaic/ floor tiles.. 6 Any other 7	5 Wall Grass/ thatch/ bamboo etc..... 1 Plastic/ polythene 2 Mud/ unburnt brick 3 Wood 4 Stone not packed with mortar.. 5 Stone packed with mortar..... 6 G.I./metal/ asbestos sheets 7 Burnt brick 8 Concrete 9 Any other 0	6 Roof Grass/ thatch/ bamboo/ wood/ mud etc... 1 Plastic/ polythene..... 2 Hand made tiles..... 3 Machine made tiles..... 4 Burnt brick..... 5 Stone..... 6 Slate..... 7 G.I./metal/ asbestos sheets..... 8 Concrete..... 9 Any other..... 0	7 Use of census house Residence..... 1 Residence-cum-other use..... 2 Shop/ office 3 School/ college etc. 4 Hotel/ lodge/ guest house etc. 5 Hospital/ dispensary etc. 6 Factory/ workshop/ workshop etc. 7 Place of worship 8 Other non-residential use..... 9 Vacant..... 0	19 Main source of drinking water Tap water from treated source 1 Tap water from un-treated source 2 Covered well..... 3 Un-covered well..... 4 Hand Pump..... 5 Tubewell/borehole..... 6 Spring..... 7 River/canal..... 8 Tank/pond/lake..... 9 Other sources..... 0	23 Type of latrine facility Flush/pour flush latrine connected to Piped sewer system 1 Septic tank 2 Other system 3 Pit latrine With slab/ventilated improved pit.. 4 Without slab/open pit..... 5 Night soil disposed into open drain.. 6 Service latrine Night soil removed by human ... 7 Night soil serviced by animals ... 8 No latrine within premises Public latrine..... 9 Open 0	26 Availability of kitchen Cooking inside house: Has kitchen..... 1 Does not have kitchen 2 Cooking outside house: Has kitchen 3 Does not have kitchen 4 No cooking 5	27 Fuel used for cooking Firewood..... 1 Crop residue..... 2 Cowdung cake..... 3 Coal/lignite/charcoal.. 4 Kerosene..... 5 LPG/PNG..... 6 Electricity..... 7 Bio-gas..... 8 Any other..... 9 No cooking..... 0
Form Number	Use only arabic numbers as indicated here 0 1 2 3 4 5 6 7 8 9 Signature of the Enumerator with Date						

Serial number	Q. 1 Name of the person start with head of household	Q. 2 Relationship to head write the relationship in full.	Q. 3 Sex Male ...1 Female 2 In case the respondent wishes to return other than code 1 or 2 then give code '3' Other...3	Q. 4 Date of birth and Age 4(a) Date of birth as per English calendar (as declared or estimated) Day- Month- Year 4(b) Age Also write age on last birthday in completed years, in box against 4(b)	Q. 5 Current marital status give code from list below	Q. 6 Age at marriage In completed years (not applicable for Never married)	Q. 7 Religion (Write name of the religion in full) Also give code in box if found in the list below For other religions, write name of the religion in full but do not give any code number	Q. 8 * Scheduled Caste (SC)/ Scheduled Tribe (ST) 8(a) Is this person SC/ST? If 'YES' give code in box SC1 ST2 If 'NO' put '3' in box 8(b) If SC or ST write name of the SC or ST from the list supplied. 8(a) 8(b)	Q. 9 Disability 9(a) Is this person mentally / physically disabled? Yes-1/No-2 9(b) If 'Yes' in 9(a), give code in the box against 9(b) from the list below 9(c) If 'multiple disability' (Code '8') in 9(b), give maximum three codes in boxes against 9(c) from the list below	Q. 10 Mother tongue write name of the mother tongue in full.	Q. 11 Other languages known write upto two languages in order of proficiency excluding mother tongue	Q. 12 Literacy status Literate ...1 Illiterate ..2 M F O	Q. 13 Status of attendance in educational institution give code from list below	Q. 14 Highest educational level attained Write the full description. For diploma or degree holder, also write the subject of specialisation.
				4(a) D D M M Y Y Y Y 4(b)					9(a) 9(b) 9(c)					
				4(a) D D M M Y Y Y Y 4(b)					9(a) 9(b) 9(c)					
				4(a) D D M M Y Y Y Y 4(b)					9(a) 9(b) 9(c)					

CENSUS HOUSEHOLD SCHEDULE-2011
PART A

Census of India 2011		Household Schedule										Confidential when filled		Use only arabic numbers as indicated here										SIDE- B	
Q. 1	Q. 15	Q. 16	Q. 17	Q. 18			Q. 19	Q. 20	Q. 21	Q. 22	Q. 23	Q. 24			Q. 25	Q. 26	Q. 27	Q. 28	Q. 29						
Serial number	Name of the person	Worked any time during last year Include part time help, unpaid work on farm, family enterprise or in any other economic activity	Category of economic activity Fill for main or marginal worker (If code '1' or '2' or '3' in Q.15)	Characteristics of workers and non-workers			Fill for marginal worker or non worker (If code '2' or '3' or '4' in Q.15)	Non-economic activity	Seeking or available for work Yes ..1 No ...2	Fill for other worker (If code '4' in Q.16) Travel to place of work 22(a) One-way distance from residence to place of work in kilometres 22(b) * Mode of travel to place of work give code from list below	Fill for person born outside this village/town Birth place if within India, write the present name of the village/town, district and state. Also write '1' for village or '2' for town in the box. if outside India, write the present name of the country and put '-' against village/town and district.	Migration characteristics			Reason for migration Work/Employment 1 Business2 Education3 Marriage4 Moved after birth 5 Moved with household6 Any other7	Duration of stay in this village/town since migration in completed years (write '00' if less than a year)	Fertility particulars								
				Fill for worker in household industry or other worker (If code '3' or '4' in Q.16) Occupation Describe the actual work	Nature of industry, trade or service where the person works/ worked or self employed. Write the full description.	Class of worker give code from list below						Place of last residence 24(a) if within India , write the present name of the village/town, district and state. Also write '1' for village or '2' for town in the box. 24(b) if outside India , write the present name of the country and put '-' against village/town and district.	24(a) if within India, write the present name of the village/town, district and state. Also write '1' for village or '2' for town in the box.	24(b) if outside India, write the present name of the country and put '-' against village/town and district.			Children surviving No. of children surviving at present	Children ever born Total no. of children ever born alive	Number of Children born alive during last one year (1st March 2010 to 28th February 2011)						
Copy from side A in same order		give code from list below	give code from list below																						
										a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country					Daughter Son	Daughter Son	Daughter Son						
										a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country					Daughter Son	Daughter Son	Daughter Son						
										a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country					Daughter Son	Daughter Son	Daughter Son						
										a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country					Daughter Son	Daughter Son	Daughter Son						

CENSUS HOUSEHOLD SCHEDULE-2011

PART B

CENSUS 2027

Houselisting and Housing Census Questions

	S No.	Question
Condition of the House	1	Line Number
	2	Building number
	3	Census house number
	4	Predominant material of Floor of the Census house
	5	Predominant material of Wall of the Census house
	6	Predominant material of Roof of the Census house
	7	Ascertain use of Census house
	8	Condition of the Census House

	S No.	Question
Amenities available to the household	17	Main source of drinking water
	18	Availability of drinking water source
	19	Main source of lighting
	20	Access to Latrine
	21	Type of Latrine
	22	Waste water outlet
	23	Availability of bathing facility
	24	Availability of kitchen and LPG/PNG connection
	25	Main Fuel used for cooking

	S No.	Question
Household Information	9	Household number
	10	Total number of persons normally residing in the household
	11	Name of the head of the household
	12	Sex of the head of the household
	13	Whether the head of the household belongs to SC/ST/Other
	14	Ownership status of the Census house
	15	Number of dwelling rooms exclusively in possession of the household
	16	Number of married couple(s) living in the household

	S No.	Question
Assets possessed by the household	26	Radio/Transistor
	27	Television
	28	Access to Internet
	29	Laptop/Computer
	30	Telephone/Mobile Phone/Smartphone
	31	Bicycle and Scooter/Motorcycle/Moped
	32	Car/Jeep/Van
Others	33	Main cereal consumed in the household
	34	Mobile number (For Census related communications only)

Expansion of Administrative Units

Census 2011–2027 Comparison

Administrative Unit	Census 2011	Census 2027
State/UT	35	36
District	640	784
Sub-district	5,990	7,092
Statutory Town	4,041	5,128
Census Town	3,892	4,580
Village	6,40,932	6,39,902

<https://www.pib.gov.in/PressNoteDetails.aspx?id=158344&NotelId=158344&ModuleId=3®=48&lang=2>



INCLASS ACTIVITY: DESIGNING SURVEY QUESTIONS AND ANALYSIS



CLASS-HOUSEHOLD SURVEY TEMPLATE

Household ID	Individual ID	Name	Sex	Age	Marital Status	Current Place of Residence- City/District	Current Place of Residence- State/Country	Place of Birth- City/District	Place of Birth- State/Country	Mother Tongue	Highest Level of Education
A	1	A1	Female	45	Married	Mumbai	MH	New Delhi	DL	Malayalam	Masters
A	2	A2	Male	47	Married	Mumbai	MH	New Delhi	DL	Malayalam	Masters
A	3	A3	Female	18	Unmarried	Mumbai	MH	New Delhi	DL	Malayalam	Secondary
B	4	B4	Female	22	Unmarried	New Delhi	DL	New Delhi	DL	Hindi	Higher Secondary
B	5	B5	Female	50	Married	New Delhi	DL	Ranchi	JH	Hindi	Masters
B	6	B6	Male	62	Married	New Delhi	DL	Chandigarh	PB	Hindi	Masters
C	7	C7	Male	52	Married	Mumbai	MH	Kolhapur	MH	Marathi	Secondary
C	8	C8	Female	42	Married	Mumbai	MH	Raigad	MH	Marathi	Upper Primary
C	9	C9	Female	17	Unmarried	Mumbai	MH	Mumbai	MH	Marathi	Secondary
C	10	C10	Female	70	Married	Mumbai	MH	Kolhapur	MH	Marathi	Upper Primary
D	11	D11	Female	18	Unmarried	Bangalore	KA	Bangalore	KA	Tamil	Secondary
D	12	D12	Female	50	Married	Bangalore	KA	Chennai	TN	Tamil	Post-Graduate
D	13	D13	Female	22	Unmarried	Bangalore	KA	Chennai	TN	Tamil	Higher Secondary



DESIGN SURVEY QUESTIONS

PREJUDICE

Prejudice differs from discrimination – prejudice is an unfavourable, preconceived opinion while discrimination is an action, making the concept of prejudice more intangible. However, we can attempt to measure it through the following ways –

(i) **Representation in media:** When we look at the communities represented in media and how they're represented, we get a fair idea of the general perception of them. When media attempt to be commercially successful, they play at the sentiments of the majority of the public. Thus, through studying and comparing commercially successful media, we can understand what said sentiments are. For example: Media about savarna characters, men, and cis het people tend to be more successful. Through a survey, we can keep a tally of these observations and come to our own conclusions.

(ii) **Individual Popularity:** Surveying individual popularity is another helpful way, in my opinion. For example, take a small village. Make a list of people, including Dalits, queer and/or trans people, women who don't conform to traditional feminine roles, and people who are none of these. Then, ask people to either rank or rate them. The results will give us a quantitative insight into the prejudices of the people surveyed.

DISCRIMINATION AGAINST GENDER AND SEXUALITY

Content:

- Discrimination is a conscious/non-conscious behavioural response of an individual towards marginalised group just on the virtue of them being that group.
- As against prejudice, discrimination is more quantifiable due to its overt nature and can be evaluated on several levels.

Methods of Evaluation:

- Comparative data analysis of historical records or trends of different gender groups in classifications such as employment rates, socio-political representation rates, legal reported records of violence and harassment, and so forth.
- Survey or Inventory Assessment of a representative sample with questions pertaining to treatment of one's own identity (by others), treatment of other identities (by oneself as well as others), and self-recognised power discrepancies.
- Media Analysis and Literature Review as pertaining to dominant rhetoric against particular genders in news and print media (popular examples would be anti-feminist and transphobic rhetoric), patterns of censorship (example, over half of US's banned book in 2023 were targeted against the LGBT+ community essentially causing an erasure of narrative), representation within media and public reception of the same, and so forth.
- Resource availability assessment which is best exemplified by availability of transitional means to offset gender dysphoria. This assessment would also entail a breakdown of public perception and conversation based around such resources.

EDUCATIONAL ATTAINMENT

Nation

- **Enrollment Ratios:** Calculate the ratio of students enrolled in different levels of education (primary, secondary, tertiary) to the total population of corresponding age groups.
- **Literacy Rate:** Measure the percentage of individuals aged 15 and above who can read and write.
- **Graduation Rates:** Measure the percentage of students who successfully complete a specific level of education (e.g., high school, college, university).
- **Educational Attainment:** Analyze the distribution of educational attainment levels in terms of completed years of education across the population.
- **Gender Disparity Index:** Assess the difference in educational attainment between males and females to identify gender-based disparities.

b. Sub-groups like religion and caste

- **Religious/Caste Literacy Rates:** Calculate literacy rates within specific religious or caste groups to identify disparities.
- **Enrollment Disparities:** Compare enrollment ratios between different religious or caste groups to identify gaps in access to education.
- **Educational Attainment by Group:** Analyze the distribution of completed years of education within each religious or caste group.
- **Gender and Group Disparities:** Study gender disparities within each religious or caste

Jahnavi Sant

WEALTH OF A HOUSEHOLD (BASED ON ASSETS AND INCOME):

- **Income Level:** Measure the annual income of a household and categorize them into income brackets (e.g., low-income, middle-income, high-income).
- **Income Distribution:** Analyze the distribution of income across the population to identify income inequality.
- **Net Worth:** Calculate the difference between a household's assets (property, investments, savings) and liabilities (debts, loans).
- **Asset Ownership:** Assess the ownership of assets like land, real estate, vehicles, and investments to gauge a household's wealth.

Descrimination:

Gender: Discrimination in gender can be measured using the aptly named gender inequality index, or GII. It uses three dimensions: reproductive health, empowerment and the labour market. It ranges from 0-1, with 0 being low inequality and 1 being high levels of inequality. Aside from that, noting down differences in popularly gendered fields— such as engineers for males and nurses for females, for example— or noticing the difference between wages for male nurses and female engineers etc.

Caste: Looking through past records of negative or criminal activities reported against minority castes and comparing those with criminal activities reported by minority castes against majority castes to see the amount of criminal reports that have been filed or followed up with for each of those castes.

Ace

Power distance in**(a) Family**

Individuals can even be asked if they feel like they have autonomy in decision making of not just themselves but even decisions that impact the entire family. How they view the authority figure in comparison with the other members of the family, is also a question that could reflect the power dynamics and how they impact the family. Dimensions of gender and age should be noted. Note the earning members of the family and whether non-earning members would respond differently if they were financially independent.

Ananya Sudarsanam -

Power Distance in Village Community Stratified by Caste and Gender

- Analyze the allocation of resources (such as education, healthcare, or community funds) among different caste and gender groups. This can provide insights into the power dynamics influencing resource distribution.
- Study the participation rates of different groups in village events, meetings, and decision-making processes. Low participation from certain groups might indicate a higher power distance for those groups.

Power Distance in Family

- Using a questionnaire with questions like who makes decisions, who speaks more, and who holds authority in various situations, and then making an analysis of the above.